Sahil Khan

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EDUCATION

Drexel University, College of Engineering | **Philadelphia, PA**
B.S. in Biomedical Engineering and M.S. in Biomechanics; Pennoni Honors College, STAR scholar June 2028
Relevant Coursework: Dynamics, Statics, Circuits, Signals and Systems, Biocomputational Modeling and Simulation, Advanced Biocomputational Languages, Advanced Programming for Engineers (C Programming)

SKILLS

Programming Languages: Python, Java, MATLAB, C/C++, JavaScript, Swift UI, Simulink, Simscape, Unreal Engine, GitLab, Linux/Unix, VS Code, Jupyter Notebook
CAD, Mechanical & Manufacturing: Siemens NX, SolidWorks, Creo, AutoCAD, Rhino, Fusion 360, Onshape, FEA, Additive Manufacturing/3D Printing, SAP, Cornerstone, Machining, GD&T
Robotics & Electrical Engineering: ROS, ESP32, Raspberry Pi, BLE, Arduino IDE, Altium Designer, KiCad, Multisim

EXPERIENCE

U.S. Naval Research Laboratory Washington, DC
Spacecraft Mechanisms Design Engineer Co-op September 2025 - March 2026
- Simulated and debugged a PID flight controller algorithm using MATLAB / Simulink, creating a Monte-Carlo simulation for parameter sweeping, validating a legacy vehicle concept for further R&D and production
- Engineered complex actuatable robotic structures constrained to high temperature and compact environments, creating mechanical test fixture assemblies using Siemens NX, various additive manufacturing processes, and machining
- Developed a Python-based application that interfaces with NX API (NX Open) to automate the generation and downloading of STEP-format CAD files, directly supporting organization wide rapid prototyping efforts for hypersonic vehicles

Estee Lauder Companies Bristol, PA
Manufacturing Engineer Co-op, Commercialization & Validation September 2024 - March 2025
- Improved product quality and reduced manufacturing costs by 15%, conducting process validation using process flow diagrams and transformative flow sheets on production lines to optimize workflows and equipment performance
- Ensured compliance across 250+ products by performing validation testing on formulations, packaging, and manufacturing processes, adhering to FDA, EMEA, and internal quality standards
- Prepared and submitted regulatory documentation via SAP and Cornerstone, keeping up with KPIs and ensuring timely approvals for time sensitive product launches worth $1M+ across global markets

Zheng Research Lab Philadelphia, PA
Independent Researcher June 2024 - December 2025
- Designed Tetra-chiral structures to improve shock absorption and lightweight strength in prosthetic legs, utilizing SolidWorks, 3D printing, ASTM-standard testing, and Finite Element Analysis (FEA)
- Analyzed mechanical properties and dimensional variations, identifying 2x2x10 tessellations as the optimal configuration for load distribution and force-to-displacement efficiency after failure mode analysis

PROJECTS

EMG-Controlled Modular Arm Prosthetic | J&J Engineering Showcase August 2025 - April 2026
- Developed an EMG signal-processing pipeline to apply Butterworth band-pass filtering, moving-window RMS smoothing, ADC conversion, rectification and MVC normalization for muscle activation detection
- Designed a modular 3D-printed prosthetic using Solidworks with pivot joints, linear actuation, alloy steel bearings, and swappable end-effector components, allowing blue collar amputees to have cost effective interchangeable components
- Engineered the embedded control and power architecture, integrating a Raspberry Pi Zero 2 W, ATmega32U4-based motor control, BMS-protected battery system, and regulated DC-DC power distribution

OpenFrame Smart-Glasses | Unity, C#, Onshape, Swift, ESP32, BLE November 2025 - Present
- Engineered a functional wearable prototype combining a CAD-designed modular frame, ESP32 firmware, BLE communication with a SwiftUI iPhone host, and an OLED–collimating lens–optical combiner display system
- Engineered a Unity-based smart-glasses simulator with custom C# scripts, real-time 3D rendering, spatial UI anchoring, input handling, and camera/FOV calibration to evaluate monocular HUD layouts and notification workflows

Agentic CAD & Multiphysics Simulation Platform | Python, React, FastAPI, CFD, FEA, ROS2 May 2026 - Present
- Developed an agentic & unified engineering platform integrating 10 analysis capabilities across structural, vibration, thermal, fluid, manufacturability, and packaging simulations enabling parametric CAD with 3D result visualizations.
- Applied the platform to a $100K+ industrial surgical endoscope development program

Drexel Formula SAE | EV Low Voltage & Manufacturing November 2023 - Present
- Designed a tractive system active light PCB in Altium Designer, assembled the ECU shutdown board, and executed brake light testing and data collection against vehicle requirements
- Designed battery transport packaging in CREO to support reliability of the high-voltage system during vehicle operations.

LEADERSHIP & INVOLVEMENT

Drexel Institute of Electrical and Electronics Engineers (IEEE) Philadelphia, PA
President / DragonHacks Lead Sep 2024 - Present
- Lead a five-member executive board in organizing weekly professional, technical, and outreach events for 700+ members
- Organized Drexel's annual 24-hour MLH hackathon for 200+ participants, designed programming and acquired sponsors